

ABSTRACT

The prevalence of hypoglycemia and jaundice in newborns necessitates early identification of complications that can impair function permanently. Complications can be serious and may include anesthetic risk, discomfort, and limited feasibility for continuous monitoring in neonates.

Current methods of diagnosing conditions involve blood collection by invasive procedures and there are limited possibilities for continuous monitoring with current techniques. In this paper, we describe the design and development of a compact, portable, low-cost and non-invasive optical sensing system to estimate levels of glucose and bilirubin in the neonatal tissue. The optical system uses multi-wavelengths of light to measure tissue absorption properties. The reflected optical signal is captured using a silicon PIN photodiode. The measured optical signal is then amplified by a transimpedance stage and converted to a digital signal using a high-resolution analog-to-digital converter.

An embedded microprocessor server (ESP32) processes the acquired optical signal data using mathematical models based on the modified Beer-Lambert Law to obtain the concentration of the biochemical being measured. A real-time digital display of processed data is provided via an OLED display. The approach described will provide reliable non-invasive point-of-care monitoring of neonates in resource-limited clinical settings.

Keywords:

Non-invasive neonatal monitoring, Optical biosensing, Multi-wavelength spectroscopy, Bilirubin detection, Glucose sensing, Modified Beer-Lambert law, Point-of-care diagnostics.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
LED	Light Emitting Diode
NIRS	Near Infrared Spectroscopy
ADC	Analog to Digital Converter
OLED	Organic Light Emitting Diode
NICU	Neonatal Intensive Care Unit
TIA	Transimpedance Amplifier
I2C	Inter – Integrated Circuits
IC	Integrated Circuits
MLO-SS	Multiplexed LED Light Sources
GPIO	General Purpose Input Output
SDA	Serial Data Line
SCL	Serial Clock Line

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND AND MOTIVATION

The timely monitoring of the health state of infants is an integral part of making sure that infants develop properly. Different diseases can develop, such as metabolic diseases including jaundice caused by high levels of bilirubin resulting in yellowness of the skin and eyes and hypoglycemia (low glucose levels in the blood). High levels of bilirubin can cause the damage to the brain tissue (kernicterus), and low glucose causes convulsions, affecting infants' development.

To monitor the levels of bilirubin and glucose in the infants' blood, conventional methods are used. Although they provide accurate data about the infants' health, conventional blood tests can cause pain to infants, making them at risk of infection. In addition, it becomes difficult to take blood from underweight infants, as there is not enough blood.

The recent advances in the area of biomedical optics and embedded sensor technology have made possible the application of new non-invasive diagnostic techniques in the medical industry. One of the techniques that make it possible to estimate biochemical parameters through optical sensing of the absorption of different light wave lengths by biological tissue is known as Near Infrared Spectroscopy (NIRS).

With the aid of NIRS, it becomes possible to differentiate between bilirubin, hemoglobin and glucose, among other molecules. It all works on the basis of the modified Beer-Lambert Law.

The main objective of this project is to develop an optical sensor for estimating the concentrations of bilirubin and glucose through a non-invasive approach. The design of the proposed device will require the integration of several LED's of different wave lengths and photodiode sensors along with a microcontroller and some other circuits for processing the output signal.

1.2 PROBLEM STATEMENT

Despite the advancements in medical technology, monitoring of bilirubin and glucose concentrations in neonates continues to be difficult. The traditional methods rely on invasive blood draws, which can be painful and potentially harmful due to infection for infants. These practices have to be done several times within the initial days of the infants' birth and thus become difficult for both infants and health practitioners.

The existing non-invasive monitors used to check the infants' bilirubin levels are unable to detect glucose concentrations in neonates. Besides, some of the non-invasive monitors are costly and need regular calibration, hence unaffordable by many healthcare facilities. Moreover, some of the existing monitoring devices give spot readings as opposed to giving continuous monitoring of the infants' levels.

Apart from that, there is a number of issues concerning the application of optical sensing in living tissue due to light scattering, absorption by tissue, and interference by other chromophores such as hemoglobin and melanin. The solution to the aforementioned issues can be obtained through complicated signal processing and calibration algorithms.

It is therefore evident that there is a demand for a more sophisticated and inexpensive platform capable of simultaneous measurement of bilirubin and glucose levels among newborns through optical sensing technology.

1.3 RELEVANCE OF THE STUDY

Jaundice and hypoglycemia are two ailments commonly experienced in infants. It is estimated that nearly 60 percent of full-term infants develop jaundice while nearly 80 percent of premature infants also suffer from jaundice, making it one of the most common problems associated with babies.

There is a lot of worry associated with the issue of hypoglycemia among infants since it is a metabolic condition, which immediately impacts a person's metabolism. This makes it

very important to diagnose the condition at its earliest so as to ensure there is no chance of developing developmental issues arising due to the neurological damage that may arise in the future.

It should be noted that there exists a novel method of monitoring that can help resolve some problems associated with blood tests. As optical techniques will be utilized, the biochemical analysis of a baby will be facilitated by light absorption through the baby's skin. Thus, it will be possible to reduce the number of blood samples required for neonatal treatment.

The suggested model includes multi-wavelength LEDs, silicon PIN photodiode, signal conditioning circuitry, and high-resolution analog to digital converter which are all operated under the control of microcontroller system.

The model will use DSP algorithms to process the signals obtained to measure the concentration of bilirubin and glucose, respectively. This model will facilitate neonate health care in that it will enable the rapid measurement of concentrations in hospital bedsides as well as improve patient comfort. The model is highly applicable to healthcare settings with limited resources.

1.4 SIGNIFICANCE OF EMBEDDED SYSTEMS

Recently, much attention has been paid to the use of optical sensors in the design of contemporary healthcare instruments due to their non-invasive nature, quick response, and high sensitivity. In view of the research concerning the absorption properties of biological tissue depending on wavelength, it becomes feasible to detect any fluctuations in the concentration of biochemicals without physically interacting with blood.

The process is based on the principles of light absorption, reflection, and scattering. Optics are extensively applied in pulse oximetry, glucose testing, and tissue oxygenation measurement. It should be noted that further developments in photonics have led to the emergence of a variety of wearable health monitoring solutions.

Optical sensing in this case relies on multi-wavelength optical illumination that

includes visible and near infrared wavelengths. The light reflected from neonatal tissue will be detected using silicon PIN photo-diode, converting it to electrical signals.

The amplification of the generated electrical signals will be done using a transimpedance amplifier. The amplified electrical signals will then be digitized through a high-resolution ADC and processed by the embedded microcontroller.

In addition, embedded processing will enable the application of digital signal processing (DSP) algorithms, such as noise reduction and calibration among others. The processed data will be transmitted to OLED display in real-time for visualization purposes.

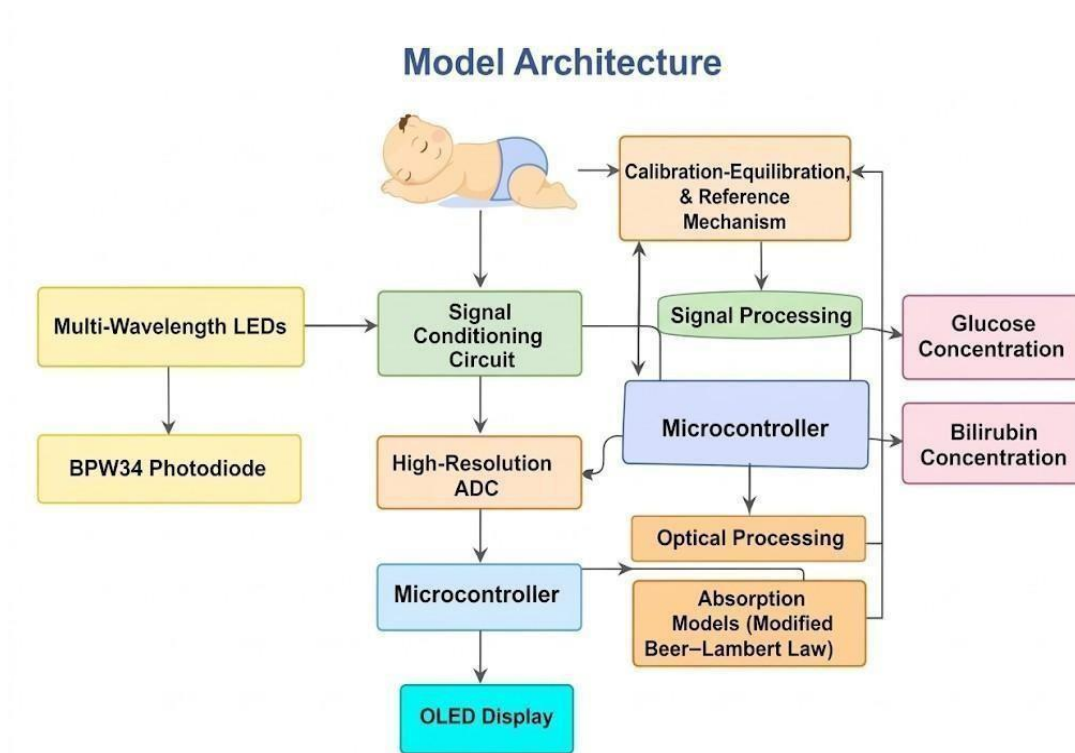


Figure 1.1 Model Architecture

1.5 SYSTEM OVERVIEW

The proposed system consists of several key components working together to detect bilirubin and glucose levels non-invasively. Multi-wavelength LEDs emit light that penetrates neonatal skin and interacts with biological tissues. The reflected light is captured by a photodiode sensor and converted into electrical signals.

These signals are amplified using a transimpedance amplifier circuit and converted into digital data using an analog-to-digital converter (ADC). A microcontroller processes the acquired signals using optical absorption models based on the modified Beer–Lambert law to estimate bilirubin and glucose concentrations. The calculated results are then displayed on an OLED screen for real-time monitoring. This integrated architecture provides a compact and portable platform capable of delivering reliable diagnostic information in clinical environments.

1.6 SCOPE AND TARGET USERS

The suggested system will address the problem of non-invasive measurement of the bilirubin level and glucose level in neonates by means of optical sensors. This system allows quick and effective measurements without causing any pain for babies. The application of this device could be very useful at hospitals, neonatal intensive care unit (NICU), and primary health-care centers, where constant measurements of the metabolic state of babies are needed. This device would be helpful for pediatricians, neonatologists, nurses, and other medical specialists in diagnosing neonatal diseases and planning treatment strategies.

This system may not only be used for clinical diagnosis but also serve for researchers working in laboratories and clinics that deal with optical sensing technologies in biomedicine and health-care for neonates. It would be helpful in the field of education for training young scientists and specialists in the area. The convenient design and portability of the system makes it suitable for remote health-care areas, thus contributing to further development of biomedical optical sensing technology.

CHAPTER 2

LITERATURE SURVEY

[1] Development and Validation of a Noninvasive Diffuse Reflectance Spectroscopic Method for Bilirubin Estimation in Neonates – Chitra et al., 2025: The study by Chitra et al. (2025) focused on the development of a non-invasive method for the estimation of bilirubin concentration in neonates using a diffuse reflectance spectroscopy technique. The proposed method for non-invasive estimation of bilirubin concentration is based on multi-wavelength optical acquisition at 450, 460, and 470 nm, focusing on the reflectance characteristics of neonates' skin. Considering the effect of chromophoric materials such as melanin and hemoglobin, the study incorporated an algorithm for chromophore interference, ensuring accurate results for bilirubin concentration estimation. The proposed method has been clinically tested with a sample population of 285 neonates, ensuring accurate correlation with standard bilirubin concentration results obtained using laboratory tests. The study confirms the feasibility of using the spectroscopy technique for effective optical sensing of bilirubin concentration in neonates, thereby supporting the concept of non-invasive optical sensing for bilirubin concentration, as implemented in the proposed optical sensing method in the current project.

[2] Non-Invasive Estimation of Hemoglobin, Bilirubin and Oxygen Saturation of Neonates Simultaneously Using Whole Optical Spectrum Analysis at Point of Care – Banerjee et al., 2023: Banerjee et al. (2023) proposed a multi-parameter non-invasive optical monitoring system that is capable of simultaneously estimating the levels of hemoglobin, bilirubin, and oxygen saturation levels in neonates. The system, termed SAMIRA, employed whole spectrum optical analysis to measure the reflected optical signals from neonatal skin tissues. A neural network-based calibration model was developed to interpret the optical spectrum and accurately estimate the physiological parameters. The proposed system was tested on 4318 neonates and showed significant correlation with the results obtained from the laboratory blood tests conducted on the neonates. The results demonstrated the potential of artificial intelligence in improving the accuracy of optical biosensors significantly. The proposed research is significant in demonstrating the potential of AI-assisted optical spectrum analysis for neonatal diagnostics, which is useful in providing insights for the proposed

intelligent signal processing techniques for bilirubin level estimation.

[3] A Terahertz Metamaterial Biosensor Using Nanocrystals for Noninvasive Urine Bilirubin Diagnosis – Valinasab et al., 2026: Valinasab et al. (2026) presented a terahertz metamaterial biosensor for the detection of bilirubin through non-invasive methods. In their paper, they proposed the use of terahertz spectroscopy, along with metamaterial sensors, for the detection of bilirubin. Nanocrystals have also been incorporated into the structure, which enhances the interaction between the sensor and the target molecules. According to the paper, the terahertz metamaterial biosensor effectively detected bilirubin through the proposed method. Moreover, the results proved that the sensor can effectively sense the bilirubin concentration. Even though the paper does not propose an approach related to the reflectance method, the paper discusses the detection of bilirubin through metamaterial sensors, which can inspire new ideas for the detection of bilirubin. Therefore, the paper is related to the topic, which is the detection of bilirubin.

[4] Transcutaneous Bilirubinometry: Serum Bilirubin Measurement Using Transcutaneous Bilirubinometer – Yamauchi et al.: In the study by Yamauchi et al., the effectiveness of the non-invasive serum bilirubin test using the transcutaneous bilirubinometry (TcB) method was assessed. The study aimed to evaluate the transcutaneous bilirubinometer device for serum bilirubin test by comparing the test results with the standard bilirubin test. The study included clinical experiments using 130 samples of neonatal blood. The correlation between the test results and the actual serum bilirubin was calculated statistically. The study concluded that the correlation coefficient was 0.99, which proved the effectiveness of the non-invasive test for serum bilirubin. However, the study also proved that the test was less effective in extreme cases of bilirubin levels. The study proved the feasibility of the optical test for the detection of jaundice in neonates. The study proved the significance of calibration in non-invasive serum bilirubin tests.

[5] Study of Effectiveness and Accuracy of Transcutaneous Bilirubinometer in Early Diagnosis of Neonatal Hyperbilirubinemia – Tanya et al., 2026: Tanya et al. (2026) conducted a clinical study to evaluate the effectiveness of transcutaneous bilirubinometers in

the early diagnosis of neonatal hyperbilirubinemia. The study was based on the early detection of neonatal jaundice using optical devices and the accuracy of the results obtained through statistical diagnostic analysis. The research paper emphasizes the significance of neonatal jaundice, as it is common in 60% of normal babies and almost 80% of premature babies. The results of the study proved the efficacy of transcutaneous bilirubinometer devices in obtaining reliable results and reducing the need for blood tests. The research validates the clinical significance of portable bilirubin detectors, leading to the development of cost-effective bilirubin detector devices for neonatal healthcare applications.

[6] Non-Invasive Optical Measurement of Neonatal Bilirubin Using Multi-Wavelength Spectroscopy – Kahn et al., 2021: Kahn et al. (2021) suggested a non-invasive bilirubin measurement technique using a multi-wavelength optical spectroscopy technique. The technique used LEDs at different wavelengths of 450 nm, 530 nm, and 660 nm to measure the characteristics of skin reflectance, which are associated with bilirubin absorption. The measured optical signals were analyzed using a spectral analysis technique, which separates bilirubin absorption from other skin components such as hemoglobin and melanin. The authors demonstrated the effectiveness of the multi-wavelength technique, which improves the accuracy of bilirubin estimation compared to single-wavelength techniques. The experimental results showed strong correlation with bilirubin measurements obtained using a laboratory setup. This reference indicates the effectiveness of optical spectroscopy in developing bilirubin non-invasive screening devices, which is essential in developing a low-cost non-invasive bilirubin monitoring system in the present project.

[7] Development of a Portable Transcutaneous Bilirubin Monitoring Device – Tayaba et al., 2020: Tayaba et al. (2020) developed a portable transcutaneous bilirubin monitoring device for the early screening of neonatal jaundice. The researchers developed a bilirubin measurement system based on the measurement of reflected light intensity from the neonatal skin surface using a blue LED illumination system and a photodiode detector. The system also used a microcontroller-based system to convert the optical signal into bilirubin concentration levels using regression models. The results were promising in terms of accuracy when the system was tested in a clinical setting and compared with the results of serum bilirubin tests. The results of the research indicated the potential of portable devices

in the early screening of neonatal jaundice, reducing the need for blood sampling procedures. The research is relevant to the proposed project because it supports the concept of implementing bilirubin detection through portable devices, which is similar to the proposed project based on the embedded system approach.

[8] Spectral Reflectance Analysis for Non-Invasive Bilirubin Detection - Bhutani et al., 2019: Bhutani et al. in their study focused on the use of spectral reflectance analysis in estimating the bilirubin level in neonates. In the study, the researchers used a spectrophotometric sensor system to detect the reflected light spectra reflected by the skin of the neonates. The researchers focused their study on the absorption of the reflected light by the bilirubin. In the study, the researchers established that the optical wavelengths used are highly sensitive to the absorption of the bilirubin. This is an interesting study that offers valuable insights into the optical parameters that can be used in the calibration of the system to improve the accuracy of the system in detecting the bilirubin level in the blood of the neonates.

[9] Machine Learning Based Prediction of Neonatal Hyperbilirubinemia – Chen et al., 2022: In the study by Chen et al. (2022), the researchers proposed the use of machine learning techniques for the prediction of hyperbilirubinemia in neonates. The study focused on the use of various algorithms, including support vector machines, random forest, and neural networks, in analyzing the features of bilirubin levels in neonatal patients. The study concluded that the use of the proposed system improved the overall accuracy of the prediction of hyperbilirubinemia in neonates. The study indicates the use of AI in the overall improvement of the accuracy of the diagnosis of the condition. This study shows the use of the system in the improvement of the overall accuracy of the diagnosis of the condition, which is useful in the development of advanced bilirubin monitoring systems.

[10] LED-Based Optical Sensor System for Transcutaneous Bilirubin Measurement – Sharma et al., 2023: In their study, Sharma et al. (2023) created an LED-based optical sensor system that is intended for transcutaneous bilirubin measurement. The device utilized multiple LEDs and photodetectors to monitor the reflectance from the skin, which is related to various wavelengths of bilirubin. The detected signal was then conditioned and

analyzed by the microcontroller, which then uses calibration models to determine bilirubin levels. The study showed satisfactory results, which correlated with bilirubin tests conducted through conventional means. In addition, the device has other benefits, which include cost-effectiveness, portability, and ease of operation, making it applicable for neonatal testing, especially in resource-limited healthcare settings. This study is applicable to the development of optical bilirubin sensors, which is related to the hardware development approach used by the proposed study.

CHAPTER 3

OBJECTIVES AND METHODOLOGY

In this chapter, the objective is to develop a non-invasive optical sensing system capable of detecting bilirubin and glucose levels in neonates. The proposed system integrates multi- wavelength optical sensing, signal conditioning circuits, and embedded processing techniques to estimate biochemical concentrations through transcutaneous optical measurements. The methodology combines optical spectroscopy principles with embedded system design to achieve accurate monitoring of neonatal metabolic parameters. The system focuses on improving diagnostic efficiency, reducing the need for invasive blood sampling, and enabling portable real- time monitoring in clinical environments.

3.1. OVERVIEW AND PURPOSE

The objective of the current design project is to develop an easy-to-use and inexpensive medical sensing device that measures the presence of bilirubin and glucose in neonates without the need for blood samples. This system works based on using multi-wavelength light emitting diodes to detect bilirubin and glucose in the neonate's skin by shining the light on the skin and detecting the reflected light through a photodiode sensor.

Various types of biomolecules have distinctive light absorption characteristics at various wavelengths. In order to estimate the concentrations of bilirubin and glucose, the absorption characteristics of the reflected light can be evaluated using optical models. The reflected light is detected by a BPW34 photodiode, amplified using a transimpedance amplifier, and digitized using a high-resolution ADC.

The detected signals are then analyzed using digital signal processing methods implemented on a microcontroller such as ESP32. The detected glucose and bilirubin amounts are then shown on an OLED display.

3.2. SYSTEM OBJECTIVES

The primary goal of this project is to create a non-invasive, compact, and portable monitoring system that can be applied to neonatal healthcare. The proposed device would estimate the level of bilirubin and glucose based on the optical properties of neonatal skin.

The first goal is to build an optical sensor system with multi-wavelength LEDs and a photodiode sensor to obtain optical signals associated with biochemical processes in neonatal skin. Such optical signals are useful in detecting and measuring the amount of bilirubin and glucose.

It is necessary to construct a signal conditioning and data acquisition system able to transform optical signals into digital numbers. To accomplish this task, one should build a transimpedance amplifier circuit and use a high-resolution ADC.

A further goal is the development of software that will employ digital signal processing methods to analyze light data and make estimates of biochemical levels through equations like the Modified Beer–Lambert Equation.

This will allow for improved precision through noise filtration and calibration of sensor outputs. The last objective is the creation of an all-inclusive system by embedding all hardware modules into one system via ESP32 microcontrollers. This will ensure a compact and efficient processing setup, allowing for effective real-time monitoring. The final objective of the project is to create an easily usable system by displaying the estimated values of bilirubin and glucose levels via OLED screens.

3.3. DESIGN AND DEVELOPMENT

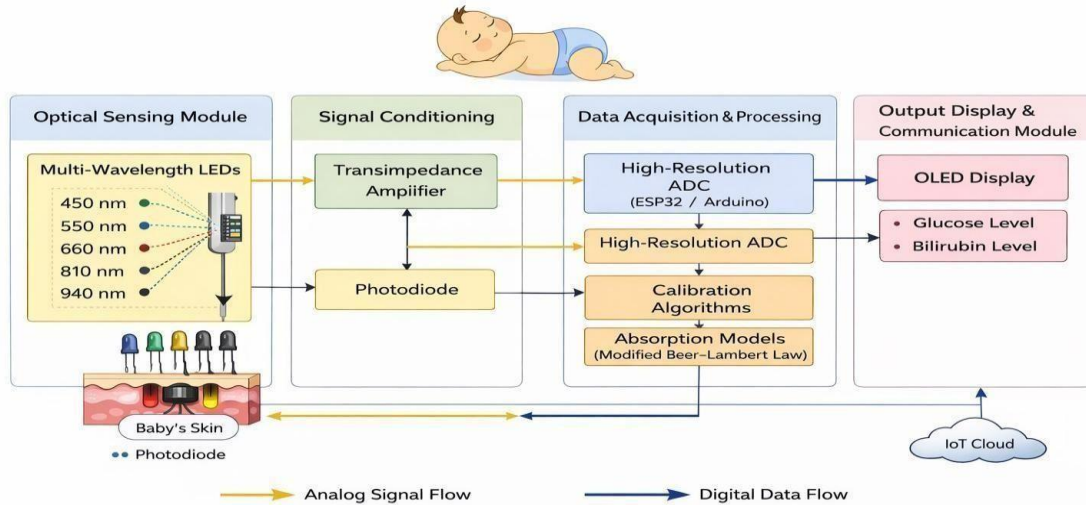


Figure 3.1 System Architecture

The system architecture follows a modular design consisting of four major components: optical sensing, signal conditioning, data processing, and output visualization.

3.3.1. Optical Sensing Module

Optical sensors are composed of multi-wavelength LEDs together with BPW34 photodiode sensors. These LEDs produce light at specific wavelengths that can react to neonatal tissues. Light waves reflected from these surfaces are captured by photodiodes and are transformed to electrical signals.

Some typical wavelengths used in the device include:

- 450 nm – Peak absorption of bilirubin
- 550 nm – Wavelength reference
- 660 nm – Absorption of hemoglobin
- 940 nm – Near-infrared wavelength for glucose and tissue absorption

3.3.2. Signal Conditioning Circuit

The electrical signal produced by the photodiode is extremely weak and requires amplification. A transimpedance amplifier circuit is used to convert photocurrent into voltage. This circuit improves signal strength and reduces noise before the signal is processed further.

3.3.3. Data Acquisition System

The analog signal which has been amplified will undergo analog-to-digital conversion using high-resolution ADC of ESP32. The ADC is connected to the microcontroller using I2C protocol.

3.3.4. Processing Unit

The role of the ESP32 microcontroller within the system is that of a central processing unit, controlling most functions of the circuit. This includes sequential switching of multicolor LEDs, collecting signals in an analog form using the ADC, and performing calculations on optical measurements collected from the photodiode. These values are computed using mathematical equations like those using the Modified Beer-Lambert law, resulting in estimated amounts of bilirubin and glucose.

3.3.5. Output Display Module

Once the values are calculated they will then transmit to an OLED display (SSD1306). The OLED display is the output device for both bilirubin and glucose readings. The use of OLED displays will yield very high quality and accurate results with high resolution on each display. To pass values from the circuit board to the microcontroller, an I²C interface will be used thus lowering the amount of wires/cables leading from the microcontroller to the circuit board.

3.4. SYSTEM IMPLEMENTATION

The process of implementing the proposed system consists of several steps. First, optimal optical wavelengths have been chosen by considering the absorbance of bilirubin and glucose molecules. Multilevel LEDs were positioned around the photodiode to make sure that the optical signals are efficiently interacting with the baby's skin. Then, the signal conditioning stage of the system has been developed using the transimpedance amplifier. Its output signals are sent to the ADC unit for further processing.

At this stage, the microcontroller sends sequential commands to the LEDs and receives corresponding sensor measurements for each wavelength. Digital signal processing methods, including filtering, normalization, and calibration are implemented for obtaining more accurate results. Finally, all the hardware components of the system are integrated into an embedded device, and the resulting data is shown on OLED display.

3.5. TESTING AND VALIDATION

In order to determine the system's efficiency, some tests have been performed. Firstly, there was hardware testing during which each separate element of the device, such as the LED light source, sensor, amplifier circuitry, converter, and controller, was being tested for proper functioning individually. Secondly, there was system testing during which all elements were connected to test their interaction with each other.

The evaluation of the system performance has been conducted by studying its reliability and accuracy during measurement processes using calibration. For that purpose, results of measurements obtained using the sensor were compared to results of standard clinical tests. User interface has been also tested regarding readability of output data on an OLED screen.

CHAPTER 4

PROPOSED METHODOLOGY

The current chapter elaborates on the methodology of the proposed development of an optical sensor system capable of measuring bilirubin and glucose concentrations in neonates. This innovative non-invasive device can be effectively used for portable, inexpensive, and risk-free diagnosis of newborns' conditions, which helps avoid dangerous procedures involving invasions.

The optical sensor works through several optical frequencies which interact with biological chromophores such as hemoglobin, glucose, and bilirubin. Optical signals obtained from neonates' skin are analyzed by the proposed system and the concentrations of bilirubin and glucose are estimated using optical absorption model based on the Modified Beer-Lambert Law. The proposed design consists of hardware modules responsible for optical signals sensing and processing as well as algorithms programmed in embedded software.

Thirty samples were examined using the proposed non-invasive optical sensor system. Bilirubin and glucose concentrations in samples were estimated on the basis of multi-wavelength optical sensing and Modified Beer-Lambert Law. The outcome of this research is illustrated in the tabulated data below, where the levels of bilirubin range from 0.3 to 1.3 mg/dL, while the levels of glucose fall between 71 and 106 mg/dL, most of which are within their respective normal values. The consistency of these readings shows that the acquisition of signal is stable and that the system works efficiently, proving that it can be used in the early detection of neonatal jaundice and hypoglycemia.

In the development of this methodology, various modules are considered, such as the Optical Sensing Module, Signal Conditioning Module, Data Acquisition and Processing Module, Output Display Module, and System Integration Module. This combination of modules functions together to ensure effective acquisition, processing, and visualization of signal.

4.1. OPTICAL SENSING MODULE

The optical sensing module forms the essential sensing device in the proposed system and is anticipated to detect optical data regarding neonatal biochemical characteristics. The optical sensing module utilizes multicolor LEDs emitting light at specific wavelengths corresponding to the absorption wavelengths of bilirubin and glucose. For example, the wavelengths employed by the system are:

- 450 nm (Blue light) - Absorption wavelength of bilirubin
- 550 nm (Green light) – Reference wavelength
- 660 nm (Red light) - Interference with hemoglobin
- 940 nm (Near-Infrared) - Absorption wavelength of glucose

When illuminated with light emitted by the LEDs, a certain amount of light will pass through the skin tissue while interacting with various chromophores, while some light will be absorbed. The light reflected back from the skin tissue will be detected using the optical sensor array. Since there are different LEDs used,

$$A_{\lambda} = \sum \epsilon_{i, \lambda} \cdot c_i \cdot L$$

This equation separates different substances using wavelengths.

4.2. SIGNAL CONDITIONING MODULE

The reflected light is measured by a BPW34 silicon PIN photodiode that produces electric current, the amplitude of which depends on the optical energy absorbed from the incoming light. Since the photodiode has the broadest sensitivity range between 430 and 1100 nm, the photodiode can detect visible and near infrared radiation used in this system. With this device, the system can measure changes in optical absorption resulting from fluctuations in the concentrations of bilirubin and glucose within neonate tissues.

$$I_{ph} = R \cdot P_{opt}$$

Where, I_{ph} = photocurrent

R = responsivity

P_{opt} = Optical Power

The electrical signal produced by the photodiode is very weak; thus, amplification is required prior to further analysis of data. Signal conditioning is performed with the help of an analog module that transforms the photoelectric current into a signal in volts. For this purpose, the current-to-voltage transformation is carried out via a transimpedance amplifier (TIA). An operational amplifier such as IC741 may be used as a basis of the TIA to provide effective amplification. Additionally, a filter can be used to reduce noise and improve signal quality.

$$V_{out} = -I_{ph} \cdot R_f$$

Where, R_f = feedback resistance

V_{out} = Output voltage

I_{ph} = photocurrent

4.3. DATA ACQUISITION AND PROCESSING MODULE

This enhanced analog signal is transformed into a digital form via a high-resolution Analog-Digital Converter (ADC) such as the controller's own ADC. This ADC supports 12-bit resolution and will be able to detect slight changes in optical signals due to biochemical concentration changes.

This processed information is then handled by the ESP32 microcontroller. The ESP32 microcontroller is responsible for several tasks in the entire process. For example, it controls the sequential illumination of LEDs. In addition, the microcontroller receives photodiode signals via the ADC, filters and normalizes the signals, and calibrates them. Finally, by using optical models based on the Modified Beer–Lambert Law, the microcontroller can calculate bilirubin and glucose concentrations.

As described above, the system uses various mathematical models based on the Modified Beer–Lambert Law. This law shows the relationship between the absorbance of light and its concentration in biological tissue. The use of several wavelengths helps separate several chromophores and measure their concentrations. As shown in the above processing step, raw optical signals are converted into biochemical measurements. Here

is how the signal was converted to concentration values.

$$C = k \cdot A + b$$

Where, C = Concentration (Bilirubin / Glucose)

A = absorbance

k = slope (sensitivity of sensor)

b = intercept (offset)

Both k and b value are collected from the research papers that are listed in the references.

4.4. OUTPUT DISPLAY MODULE

Healthcare professionals will be able to easily read calculated biochemistry results outputted by the biochemical analyzer via an output display module. Medical personnel will see the values of the calculated bilirubin and glucose levels in real-time on the OLED (SSD1306) display module.

Because of the characteristics of an OLED display including high contrast ratio, low power requirements, and small physical size; the OLED display works well as part of a portable biomedical device. The OLED display communicates with the microcontroller via the I²C interface enabling quick and easy transfer of data and simplified wiring

The OLED display can also provide other types of system information in addition to displaying the measured values. This includes real-time updates to the measured values from the sensors, warnings to the operator when the measured values exceed a predetermined neonatal range, and sensor status messages. By providing a real-time visual display of a neonate's health status, a healthcare practitioner will be able to make an accurate and prompt assessment of the neonate's health status.

4.5. SYSTEM INTEGRATION AND EMBEDDED CONTROL MODULE

The system integration block incorporates all the hardware and software components into one functional unit. The microcontroller is used as the control device, controlling the activities of the LED diodes, sensors, signal processing circuits, and display circuitry.

The operation starts with sequential activation of the multi-wavelength LEDs by the microcontroller. The optical signals obtained from the reflected light detected by the photodiode undergo amplification and digitization by the ADC component. The microcontroller further processes these signals using calibration coefficients and models of optical absorption, obtaining the level of bilirubin and glucose.

As the final step, the results of the processing are shown on the OLED display. This enables real-time monitoring of neonatal physiological parameters. Wireless communication technology can be added in order to provide remote diagnosis.

4.6. SYSTEM WORKFLOW AND OPERATIONAL PROCESS

Continuous measurement of biochemical parameters through real-time monitoring of neonates will be accomplished by a neonatal monitoring device. This device first turns on multiple wavelengths of LEDs that illuminate the neonates' skin. The reflected light from the skin is detected by the photodiode and converted into an electrical signal. The signal is then amplified by the signal conditioning circuitry and converted into digital data using an ADC.

The microcontroller will process the digital data using calibration algorithms and optical absorption models to determine the concentrations of bilirubin and glucose present. The calculated data will then be displayed on an OLED screen for real-time monitoring.

Although this system can function automatically, it is intended to assist healthcare professionals and not replace their clinical judgment. Healthcare providers will be able to utilize the monitoring system as a decision-support device that will enable them to easily monitor for abnormal glucose or bilirubin levels and take action as necessary.

CHAPTER 5

RESULTS AND DISCUSSION

5.1. PERFORMANCE ANALYSIS

5.1.1. Signal Stability

Using both median filtering and averaging techniques, noise from the photodiode signal was greatly reduced. This allowed for accurate intensity readings, even when there was slight movements in the patient's fingers and from the surrounding environment.

5.1.2. Multi-Wavelength Effectiveness

Each of the LEDs provided a different wavelength with different properties for selective absorption. The blue and green wavelengths are more sensitive to the amount of bilirubin in the blood. The infrared and red wavelengths correspond more closely to optical variations resulting from changes in glucose levels. This supports the use of multi-spectral sensing for the estimation of biochemical substances.

5.2. HARDWARE PERFORMANCE

The system's hardware performance was found to be dependable and useful. The ESP32-based data acquisition and the ADC gave steady and reliable readings of analog signals. The OLED screen made it easy to see the measured parameters in real time, which made the system easy to use. The small breadboard-based implementation also shows that it is possible to make a portable and cheap device for monitoring newborns.

5.3. INTERPRETATION

The optical sensor unit was assessed using the finger placed on the LED and photodiode module and analyzed for thirty seconds. Within the thirty seconds of data acquisition, the ESP32 was able to provide four colors, including blue, green, red, and infrared colors. The data was then presented through the OLED screen after performing all mathematical calculations. The same individual performed multiple calibrations of the API, indicating that the section responsible for collecting data using an optical sensor and

processing it worked correctly. Lastly, the numeric OLED screen implies that the optical sensor worked correctly.

Table 5.1. Table of readings

S.No	Sample	Gender	Age	Bilirubin (mg/dL)	Glucose (mg/dL)
1.	Sample 1	F	21	1.2	75
2.	Sample 2	F	20	0.9	80
3.	Sample 3	M	21	1.1	101
4.	Sample 4	F	21	0.4	91
5.	Sample 5	M	21	0.7	78
6.	Sample 6	F	20	1.3	84
7.	Sample 7	F	20	0.9	79
8.	Sample 8	M	20	1.2	87
9.	Sample 9	F	21	0.8	94
10.	Sample 10	M	20	0.3	105
11.	Sample 11	M	19	0.5	83
12.	Sample 12	M	19	0.6	79
13.	Sample 13	F	19	0.7	92
14.	Sample 14	M	21	1.1	80
15.	Sample 15	F	21	0.7	86
16.	Sample 16	F	19	1.2	95
17.	Sample 17	F	21	0.4	77
18.	Sample 18	M	20	0.9	82
19.	Sample 19	F	21	0.8	78
20.	Sample 20	M	19	1.2	94
21.	Sample 21	F	21	0.6	76
22.	Sample 22	F	21	1.0	89

23.	Sample 23	M	20	0.5	96
24.	Sample 24	F	19	0.7	80
25.	Sample 25	M	19	1.2	106
26.	Sample 26	F	20	0.3	89
27.	Sample 27	F	21	0.7	72
28.	Sample 28	M	19	1.1	93
29.	Sample 29	F	19	0.4	71
30.	Sample 30	M	20	0.9	99

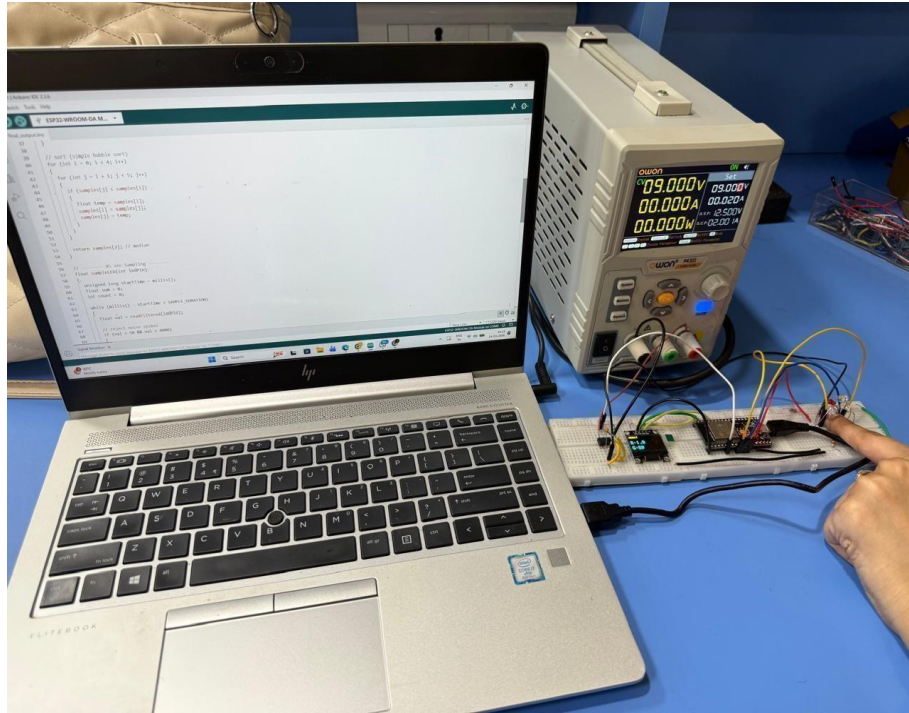


Figure 5.1. Experimental setup

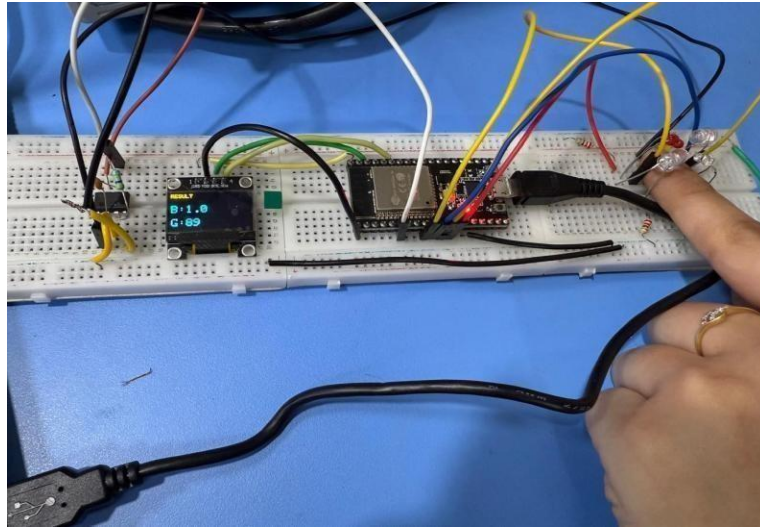


Figure 5.2. Measurement of glucose and bilirubin

5.4. DISCUSSION

This system works on the optical absorption principle which states that different wavelengths will behave differently when interacting with biological tissues. A change in intensity of detected light from tissue was used to create absorbance with a logarithmic calculation in order to provide useful biochemical data. Several LEDs could be used to determine the different absorption characteristics of bilirubin and glucose. By calculating the relative difference between selected pairs of wavelengths, and measuring the results, it was possible to isolate and accurately estimate the individual parameters. The application of signal processing methods such as filtering and averaging has helped improve the stability of the readings through the reduction of noise and fluctuation. The sequential activation of LEDs has also assisted in ensuring that each measured wavelength was measured independently from other wavelengths; These results suggest that the system can do real-time optical measurements and convert those measurements into interpretable biomedical parameters. The reliability of the output and smoothness of the display suggests that the algorithm, hardware interface and data processing pipeline are all functioning together seamlessly. Due to ethical constraints and limited access to neonatal subjects, the experimental evaluation was conducted on adult participants. The obtained results are used for system validation and may vary when applied to neonates due to differences in physiological and optical properties.

CHAPTER 6

CONCLUSION

This project has demonstrated that Multicolour Light Detection using Multiplexed LED Light Sources (MLO-SS) and the use of a Photodiode as the detecting device are cost effective, simple means to optically detect multiple colours of light incident with a sample of biological tissue utilizing Time-Multiplexed Lighting of the LEDs and Photodiode-based Signal Acquisition.

The TIA (Trans Impedance Amplifier) was constructed utilising an IC741 operational amplifier. This amplifier was selected due to its ability to convert the low-level current generated by the photodiode into a measurable voltage. The limitations of the IC741 with low voltage applications will affect the performance of the final prototype but will be sufficient for this project as long as sound circuit design is adhered to. Additionally, the simplicity of the TIA design will be improved with the incorporation of the built-in analogue to digital converter (ADC) of the ESP32 microcontroller due to the elimination of additional components with the ability to acquire enough resolution of the signal to perform an analysis.

The Time-Multiplexed approach to the measurement of the LED/Photodiode combination allowed each wavelength of light to be separated thus eliminating the effects of common optical interference improving the reliability of measurements. To ensure that the output readings of the sensors were reliable and stable, appropriate grounding and systematic testing of the sensors were performed.

Once reliable performance was established; variations in the output of the sensors due to the arrangement of the LEDs or the configuration of the finger to the photodiode, were produced and monitored. Ultimately, this project suggests that a low-cost, portable optical sensing system is feasible.

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APPENDICES

8.1. APPENDIX A – PROJECT OVERVIEW

8.1.1. Circuit Diagram Description

The schematic diagram gives a visual representation of how the components communicate to produce a functional device. Each component has a unique purpose and utilizes the ADC input on the ESP32 for receiving data; this information displays on the OLED screen. The transimpedance amplifier increases the intensity of light at the photodiode sensor, allowing one to know if the photodiode has received enough light. All four LEDs are controlled by the ESP32 using GPIO pins and supply enough light to the object being illuminated, then reflecting back onto the photodiode, enabling you to capture the reflected light at different times. The OLED screen makes it simple to observe the vs. amplitude of the measurement taken at the photodiode. The system's power source is a regulated source that allows for consistent performance with the ESP32.

8.1.2. Algorithm Summary

To set up the system and be able to run, you must first connect all components (ESP32 microcontroller, OLED Display and ADC Module). Next, you will light up LEDs individually at the end of your finger to allow light to pass through your tissue. The light returned from your finger is measured with a photodiode and the average of those measurements will be calculated for accuracy and to reduce noise. The absorbance of the light that was detected will be calculated using the Beer-Lambert Law. Once the absorbance calculations are complete, they will be used as an input to extract the features needed for bilirubin & glucose and modify the features for OLED display in real-time.

8.1.3. Assumptions and Limitations

The purpose of this system is to estimate blood sugar and bilirubin levels based on what the user uses the attachment to their body, but not necessarily measuring what would be

measured clinically to determine blood sugar or bilirubin levels. People's skin reflects light differently, leading to their measurement of the reading having a large degree of error. The skin temperature can also be different than the ambient temperature therefore affecting all readings as well. The system must undergo calibration processing using actual data from patients (i.e., clinical datasets) before it is acceptable for clinical usage.

8.1.4. Components List

Table 8.1. Components list

S.No.	Component Name	Specification	Quantity
1.	ESP32 Wroom 32 Board	Wifi + ADC	1
2.	Blue LED	470 nm	1
3.	Green LED	525 nm	1
4.	Red LED	660 nm	1
5.	IR LED	940nm	1
6.	Photodiode	BPW34	1
7.	Op - Amp	LM741	1
8.	OLED display	0.96" SSD1306 I2C	1
9.	Resistors	Various Values	As required
10.	Breadboard & Wires	-	As required

8.1.5. References (Basic)

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8.2. APPENDIX B - INDIVIDUAL CONTRIBUTIONS

JANASRUTHIKA P R:

- Developed the architecture of the device that measures glucose and bilirubin levels through absorption of light using correct optical components (multi-wavelength LEDs and photodiodes).
- Created the circuitry to amplify the weak signal from the photodiode
- Connected both the transimpedance amplifier and photodiode to the ESP32
- Helped in the experimental set up of the circuit and made adjustments to improve signal quality and reduce noise.
- Constructed a display viewed on the OLED screen providing real-time data and alerting the user through the OLED screen of device issues.

JEEVIKA HARSHINE S:

- Created embedded software for the ESP32 to control LED switching
- Retrieve readings from the ADC of the reflections from multi-wavelength light from two different sensors.
- Utilized several types of signal processing methods to enhance the accuracy of the readings
- Utilized the Modified Beer-Lambert Law to mathematically calculate and/or estimate concentrations of glucose and bilirubin.
- Documented all work performed in order to calibrate, test/verify the functionality of the system and to produce technical documentation required to support this effort.

8.3. APPENDIX C – PROJECT OUTCOME





8.4. PLAGARISM REPORT

Report_S6			
ORIGINALITY REPORT			
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